Modeling Sensitivity to Red and Green Small Spots: The Effect of Cone Topography and Spectral Sensitivity

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Background

If the detection of small spots on a neutral background is based on the sum of equally weighted L and M-cone signals, then detection should depend on the numbers of L and M-cones stimulated.

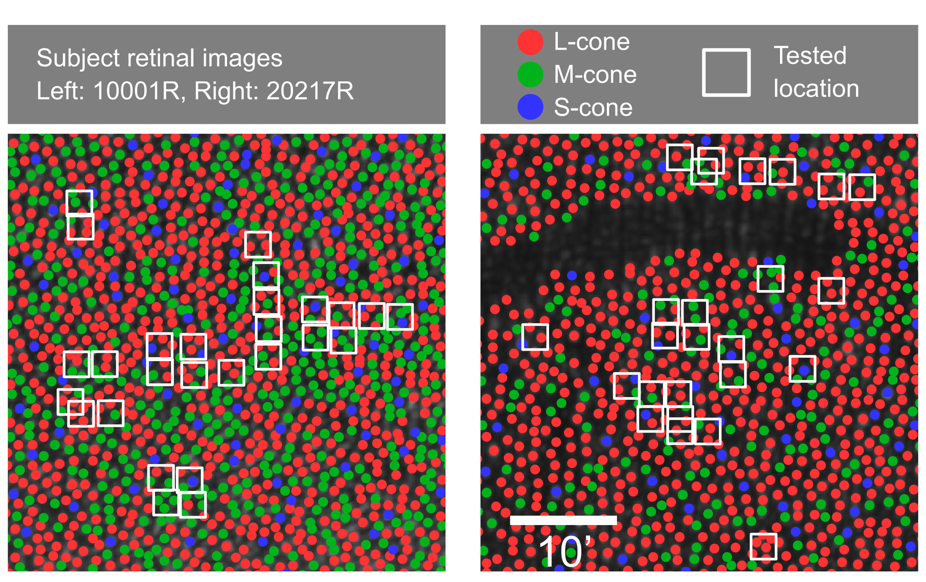
We determined detection thresholds for adaptive optics-corrected red (680 nm) or green (543 nm) flashes as a function of local L:M ratio, in subjects whose cones were spectrally classified using optoretinography.

We compared the ratio of 680 nm to 543 nm sensitivity (S_{680}/S_{543}) at the different L:M ratios tested with the predictions of a simple additive model, and with ideal observer-derived thresholds.

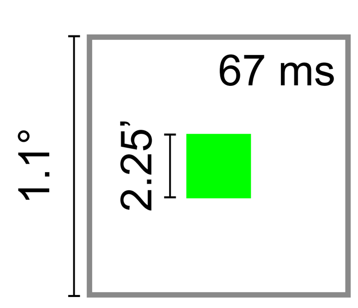
Methods

Sensitivity measurements

Subjects



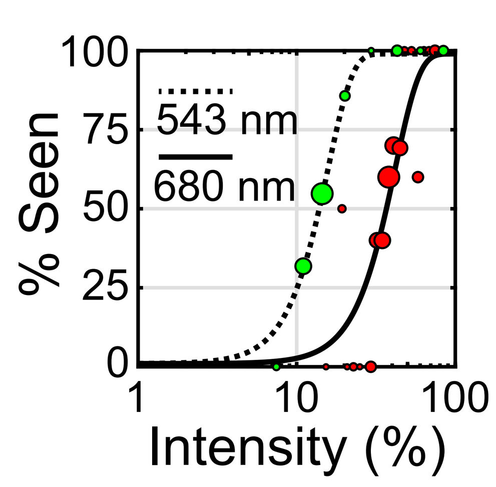
Stimuli



Test:
λ = 543 or 680 nm

Background:
(white, 1700 cd/m²)

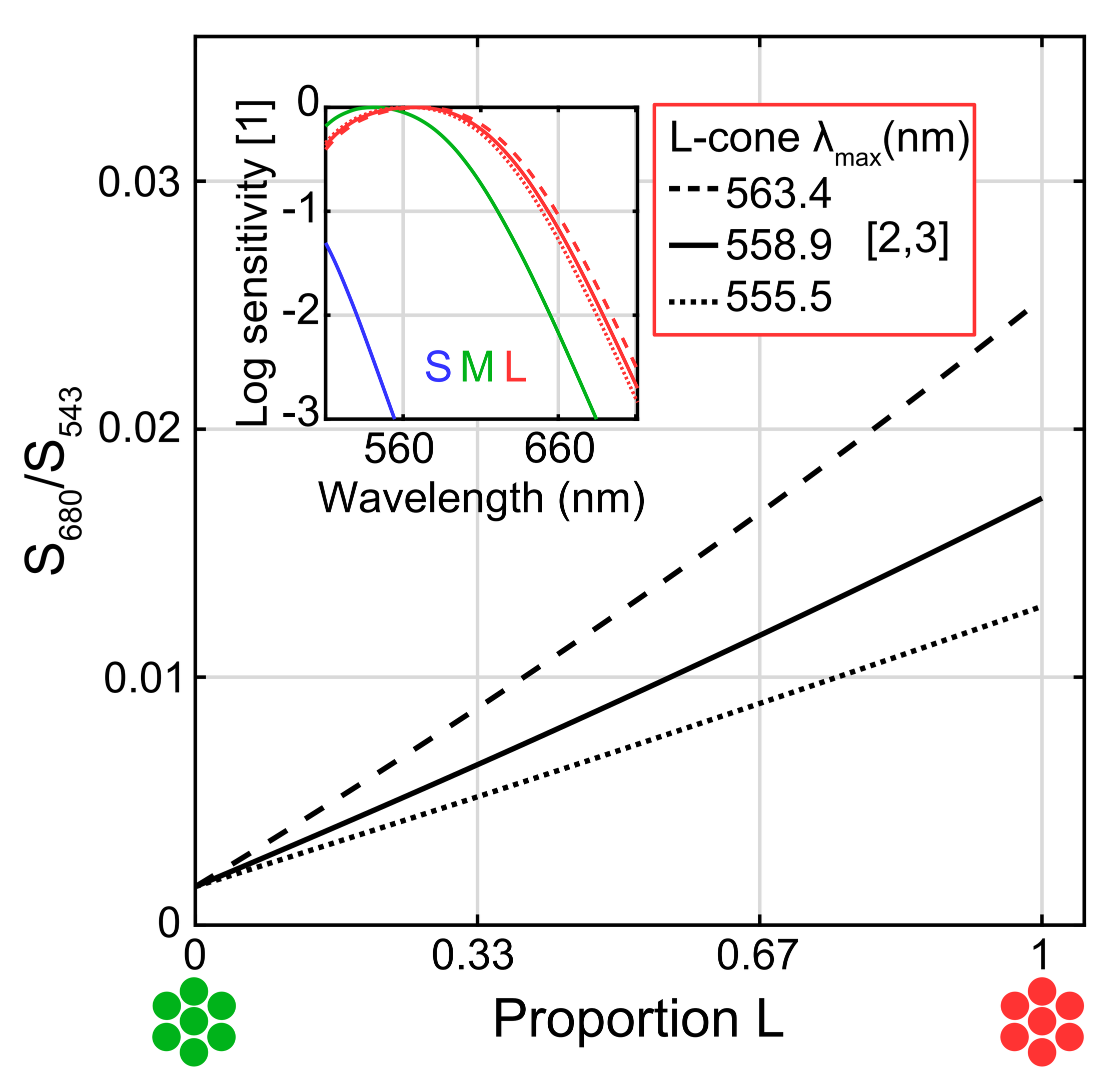
Task



Seen / not seen

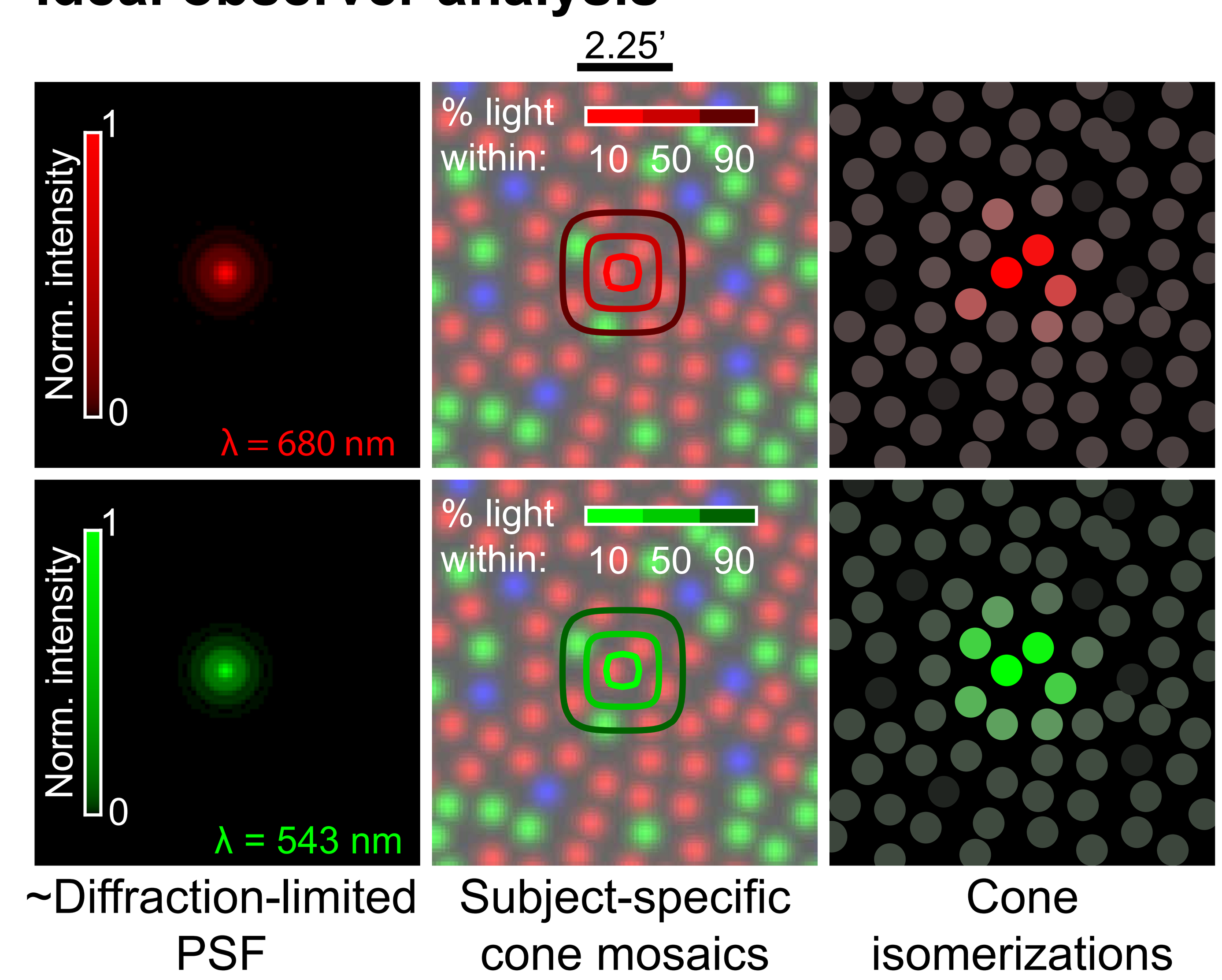
Sensitivity predictions

Additive model



Sensitivity =
L-cones x L-cone sensitivity to test light
+ # M-cones x M-cone sensitivity to test light

Ideal observer analysis



$$d' = \frac{\sum_{i=1}^{n} (\beta_i - \alpha_i) \ln(\beta_i/\alpha_i)}{\left[0.5 \sum_{i=1}^{n} (\beta_i + \alpha_i) \ln^2(\beta_i/\alpha_i)\right]^{1/2}}$$

n: number of cones
α_i: mean background isomerizations in i^{th} cone
β_i: mean background + test isomerizations in i^{th} cone
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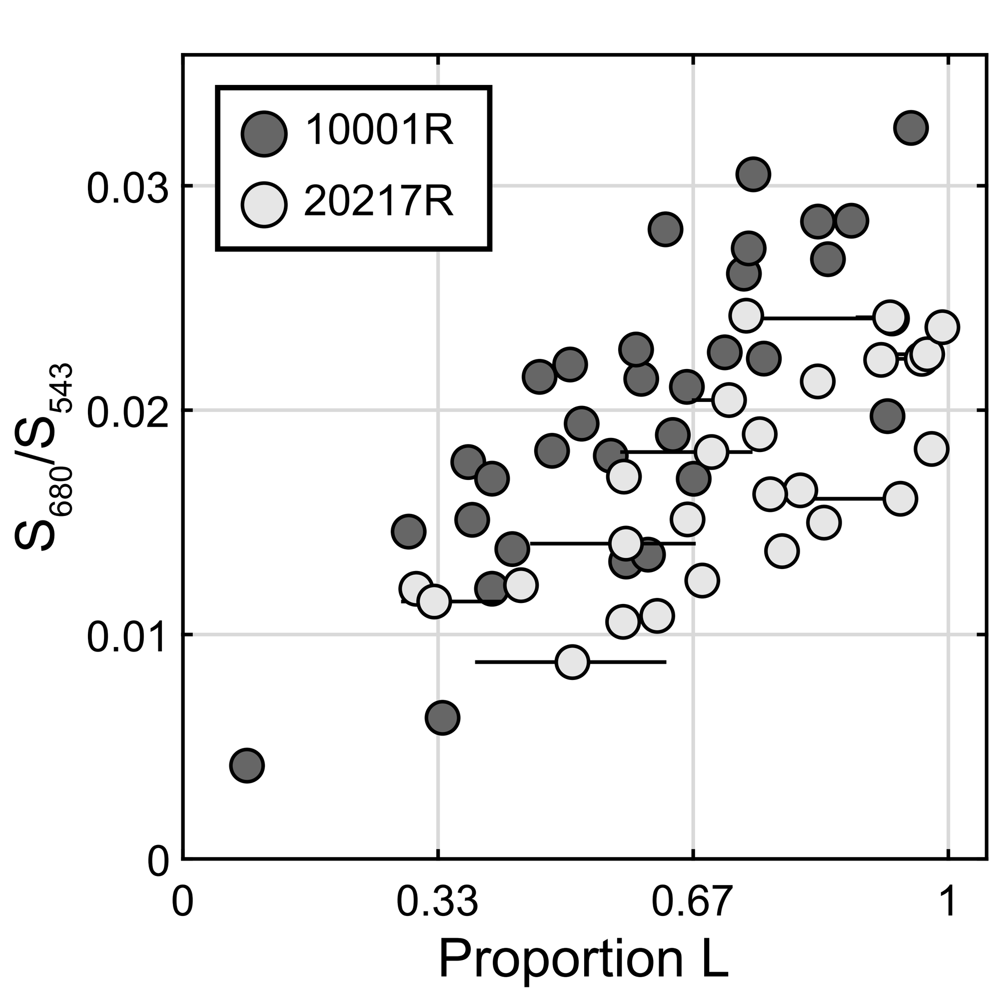
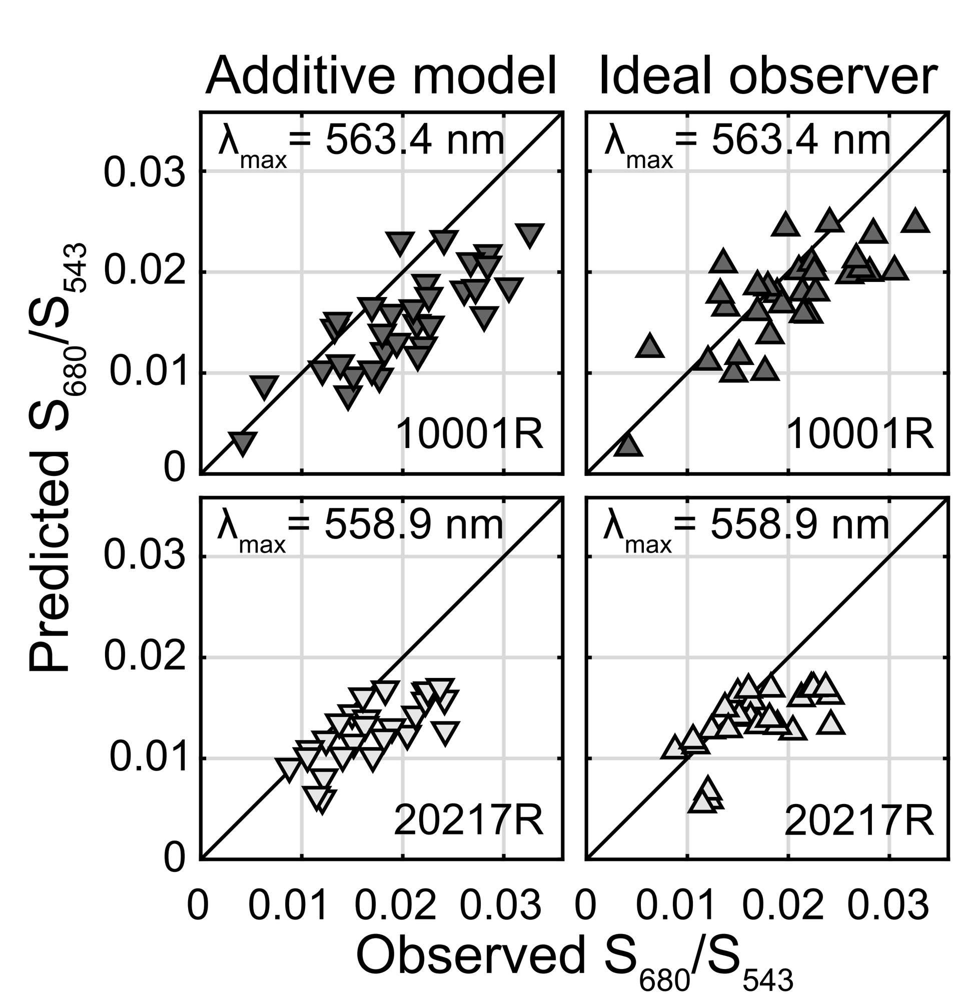
Results and Conclusions

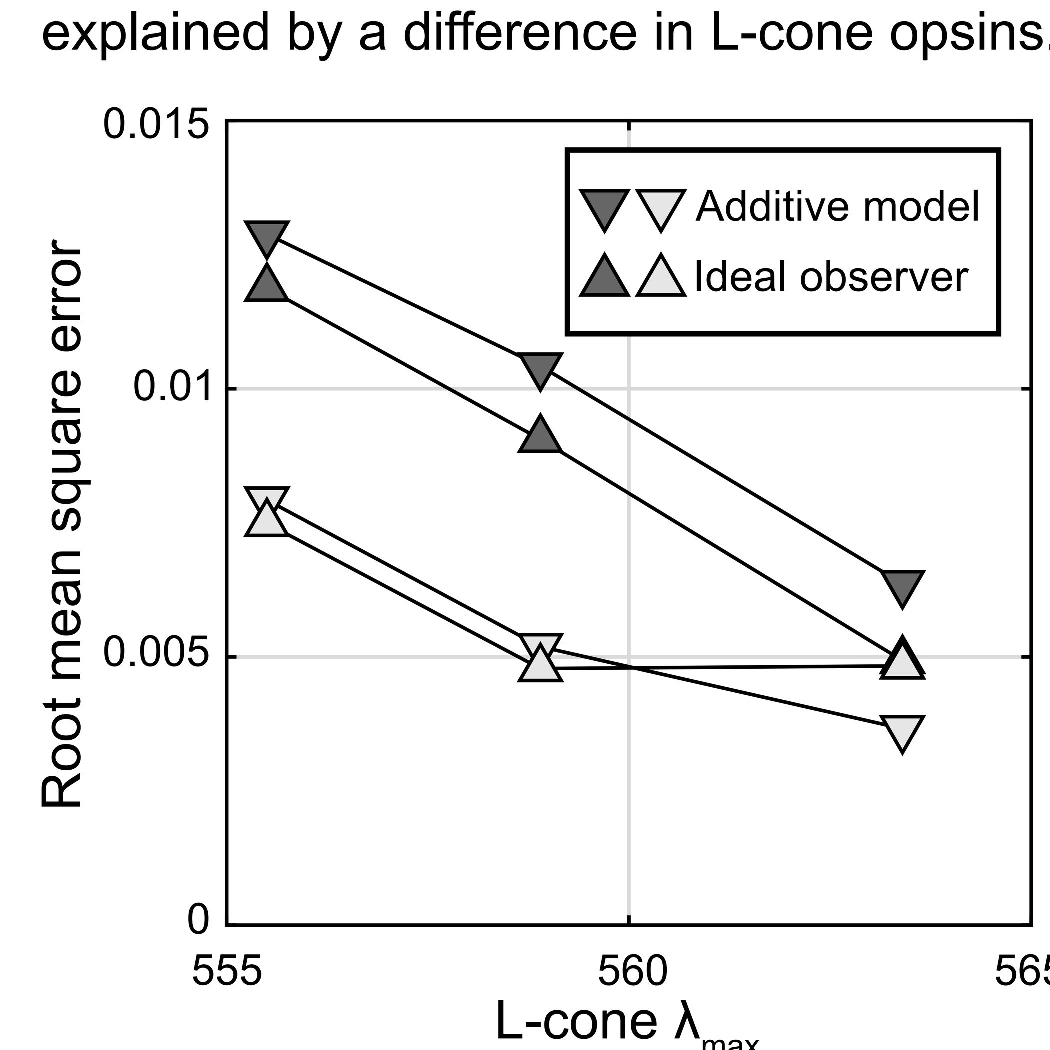
1. Relative sensitivity (S_{680}/S_{543}) to long-wavelength small spots increased with the relative number of L-cones stimulated (proportion L).



2. Our results were similar to predictions made by an ideal observer at the level of cone isomerizations, and by a simpler cone-additive model.



3. The relationship between prediction error and the assumed L-cone spectral peak varied between subjects, possibly explained by a difference in L-cone opsins.



References
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